

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT _*CO-SC(MODEL-A&APEX) Date: 19-03-26
Time: 3 Hrs Max. Marks: 300

19-03-26_SR.IIT_STAR CO-SC(MODEL-A&APEX)_JEE MAIN_GTM-10_SYLLABUS

PHYSICS: TOTAL SYLLABUS

CHEMISTRY: TOTAL SYLLABUS

MATHEMATICS: TOTAL SYLLABUS

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PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. In an experiment to determine the acceleration due to gravity (g), the period of a simple pendulum is given by the formula $T = 2\pi\sqrt{\frac{L}{g}}$. A student measures the length of the pendulum (L) to be 50.0 cm using a meter stick with an accuracy of 2 mm. The time for 50 oscillations is recorded as 100 s using a stopwatch with a resolution of 1 s. What is the maximum percentage error in the determination of g from this experiment?

A) 2.0% B) 2.4% C) 2.8% D) 4.4%

2. Given below are two statement: one is labeled as Assertion A and the other is labelled as Reason R:

Assertion A: For an ideal gas undergoing an adiabatic expansion, the magnitude of the slope of the P-V curve is greater than that of an isothermal expansion at the intersection point of the two curves.

Reason R: The ratio of specific heat at constant pressure to specific heat at constant volume is always greater than 1.

In the light of the above statements, choose the correct answer from the options given below:

A) A is true but R is false.

B) A is false but R is true.

C) Both A and R are true and R is the correct explanation of A.

D) Both A and Rare true but R is not the correct explanation of A.

3. Match List-I (physical quantities related to satellite orbiting a planet at a distance R from center) with List-II (their dependence on R).

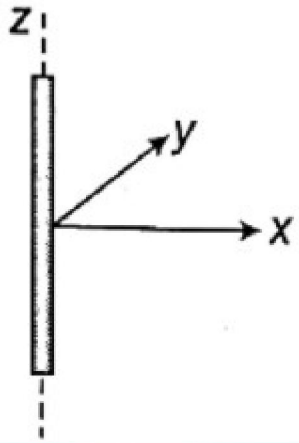
	List-I		List-II
A)	Orbital velocity	i)	$\propto R^{3/2}$
B)	Time Period of Revolution	ii)	$\propto R^{-1}$
C)	Angular momentum about centre	iii)	$\propto R^{-1/2}$
D)	Gravitational Potential Energy	iv)	$\propto R^{1/2}$

A) A-III, B-I, C-IV, D-II B) A-II, B-III, C-I, D-IV

C) A-I, B-II, C-III, D-IV D) A-III, B-I, C-II, D-IV

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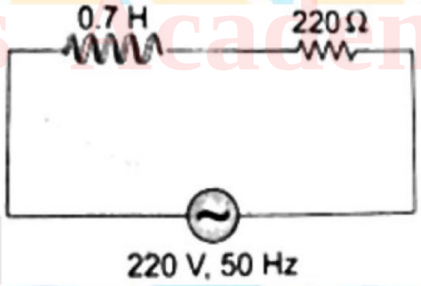
4. An infinitely long wire is kept along z-axis from $z=-\infty$ to $z=+\infty$, having uniform linear charge density $\frac{10}{9} nC/m$. The electric field at point (6 cm, 8 cm, 10 cm) will be



A) $\left(120\hat{i}+160\hat{j}+200\hat{k}\right)$ B) $200\hat{k}V/m$

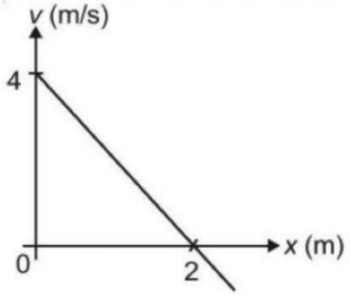
C) $\left(160\hat{i}+120\hat{j}\right)V/m$ D) $\left(120\hat{i}+160\hat{j}\right)V/m$

5. A coil has an inductance of 0.7 H and is joined in series with a resistance of 220Ω. When an AC e.m.f. of 220 V, 50 Hz is applied to it, find the wattless component of current in the circuit is



A) 0.2 A B) 0.4 A C) 0.5 A D) 0.7 A

6. The velocity (υ) of a particle moving along x-axis varies with its position x as shown in figure. The graph of acceleration (a) of particle with position (x) is a



A) Parabola B) Circle

C) Rectangular hyperbola D) Straight line

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7. An atom ${}^{8}_{3}X$ is bombarded by shower of fundamental particles and in 10s, this atom absorbed 10 electrons, 10 protons and 9 neutrons. The percentage growth in the surface area of the nucleus is recorded by

A) 250% B) 125% C) 225% D) 150%

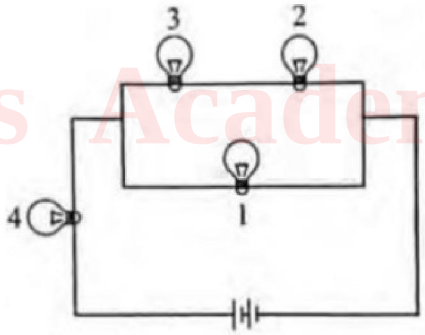
8. In Young's double slit experiment the intensity at a point is $(3/4)^{th}$ of the maximum intensity. The possible angular position of this point is

A) $\sin^{-1}\left(\frac{\lambda}{3d}\right)$ B) $\sin^{-1}\left(\frac{\lambda}{2d}\right)$ C) $\sin^{-1}\left(\frac{\lambda}{6d}\right)$ D) $\sin^{-1}\left(\frac{\lambda}{4d}\right)$

9. A particle of mass m is projected obliquely from ground such that the maximum and minimum de-Broglie wavelengths are λ_1 and λ_2 respectively. The difference in the maximum and minimum kinetic energy of the particle is

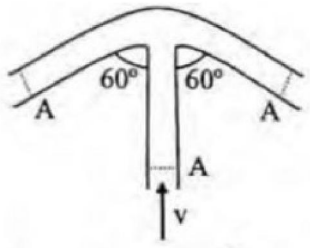
A) $\frac{h^2}{2m}\left[\frac{1}{\lambda_1^2}-\frac{1}{\lambda_2^2}\right]$ B) $\frac{h^2}{2m}\left[\frac{1}{\lambda_2^2}-\frac{1}{\lambda_1^2}\right]$ C) $\frac{h^2}{2m}\left[\frac{1}{\lambda_2}-\frac{1}{\lambda_1}\right]$ D) $\frac{h^2}{2m}\left[\frac{1}{\lambda_1}-\frac{1}{\lambda_2}\right]$

10. All bulbs consume same power. The resistance of bulb 1 is 36Ω. The resistance of the bulb 4 is



A) 4 ohm B) 9 ohm C) 12 ohm D) 18 ohm

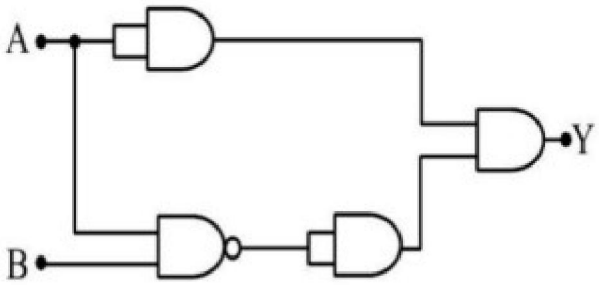
11. An ideal liquid of density ρ is pushed with velocity v through the central limb of the tube shown in the figure. What force does the liquid exert on the tube? The cross sectional areas of the three limbs are equal to A each. Assume stream-line flow.



A) $\frac{9}{8}\rho Av^2$ B) $\frac{5}{4}\rho Av^2$ C) $\frac{3}{2}\rho Av^2$ D) ρAv^2

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12. Identify the correct truth table of the given logical circuit.



A)

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

B)

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

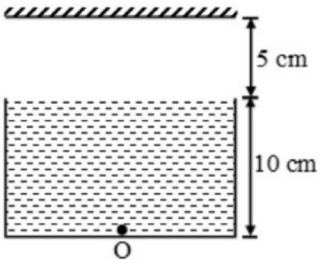
C)

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

D)

A	B	Y
0	0	0
0	1	0
1	0	1
1	1	0

13. Consider the situation shown in figure. Water $\left(\mu_w = \frac{4}{3}\right)$ is filled in a beaker upto a height of 10 cm. A plane mirror is fixed at a height of 5 cm from the surface of water. Distance of image from the mirror after reflection from it for an object O at the bottom of the beaker is



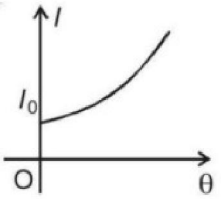
(1) 15 cm (2) 12.5 cm (3) 7.5 cm (4) 17 cm

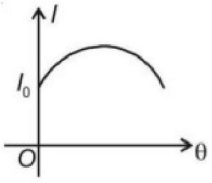
14. Two particles of mass "p" and "q" $(q < p)$ are at a certain distance apart. On interchanging the position of the masses, the centre of mass of the system is found to shift through a distance "d". The distance between the two particles is

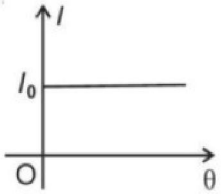
A) $\frac{d(p-q)}{q}$ B) $\frac{d(p-q)}{p}$ C) $\frac{d(p+q)}{p-q}$ D) $\frac{d(p-q)}{p+q}$

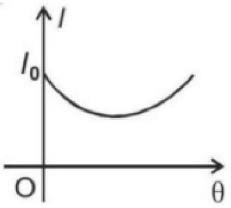
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15. A uniform square plate has a moment of inertia I_0 about an axis lying in its plane, passing through its centre and making an angle θ with one of the sides. Which graph represents the variation of I with θ?

A) 

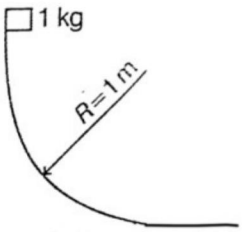
B) 

C) 

D) 

16. A block of mass 1 kg slides down a curved track i.e. one quadrant of a circle of radius 1m. If the coefficient of friction between the block and the surface is 0.5. If the velocity of block is $\sqrt{10}\,m/s$ at the end of the track then find the work done by frictional force upon reaching the bottom is

(take g = 10 m/s^2)



A) $-7.85J$ B) $-15.7J$ C) $-10J$ D) $-5J$

17. A metallic sphere of radius R_1 is charged to a potential V. If it is enclosed symmetrically by a spherical conducting shell of radius R_2 and then connected to it by a thin conducting wire. Now, the potential of sphere will be

A) zero B) $V\left(\frac{R_1}{R_2}\right)$ C) $V\left(\frac{R_2}{R_1}\right)$ D) $V\left(\frac{R_1}{R_1+R_2}\right)$

18. The relation between internal energy U, pressure P and volume V of gas in an adiabatic process is $U=a+bPV$, where a and b are positive constants. The value of molar specific heats ratio (γ) is

A) $\frac{a}{b}$ B) $\frac{b+1}{b}$ C) $\frac{a+1}{a}$ D) $\frac{b}{a}$

19. A pipe of length 85 cm is closed from one end. Find the number of possible natural oscillations of air column in the pipe whose frequencies lie below 1250Hz. The velocity of sound in air is 340 m/s

A) 4 B) 12 C) 8 D) 6

For More Join: @IITJEE_Advanced

20. Two wires of same length are shaped into a square and a circle. If they carry same current, ratio of the magnetic moments (square to circle) is____

A) $2:\pi$ B) $\pi:2$ C) $\pi:4$ D) $4:\pi$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

21. Hydrogen atom is excited from ground state to another state with principal quantum number equal to 4. Then the number of spectral lines in the emission spectra will be:

22. In a compound microscope, the focal lengths of objective and eye lenses are 1.2 cm and 3 cm, respectively. If the object is put 1.25 cm away from the objective lens and the final image is formed at infinity, the magnifying power of the microscope is (Take, D = 25cm)

23. Acceleration of a particle moving rectilinearly as $a=(4-2x)m/s^2$. It is at instantaneous rest at $x=0$. At what position X, will the particle again comes to instantaneous rest? (in centimeters)

24. A beam of unpolarized light of intensity I_0 is passed through a polaroid A and then through another polaroid B which is oriented so that its principal plane makes an angle of $45°$ relative to that of A. The intensity of the emergent light is $\frac{I_0}{x}$, then 'x' is_____

25. A moving coil galvanometer allows a full scale current of 10^{-4} A. A series resistance of $30\,K\Omega$ is required to convert the above galvanometer into a voltmeter of range 0 – 5V. Therefore the value of shunt resistance required to convert the above galvanometer into an ammeter of range 0 -10 mA is_____ (to the nearest integer in ohm)

For More Join: @IITJEE_Advanced

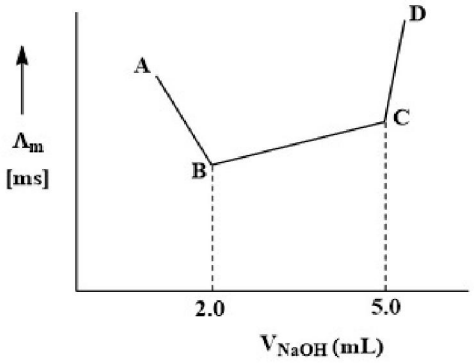
CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. 40 mL of a mixture of CH_3COOH and HCl (aqueous solution) is titrated against 0.1 M NaOH solution conductometrically. Which of the following statement is correct?



I) The concentration of CH_3COOH in the original mixture is 0.0075

II) The concentration of HCl in the original mixture is 0.005M

III) HCl is neutralized first followed by neutralisation of CH_3COOH

IV) Point 'C' indicates the complete neutralization HCl

A) I ,II, IV B) I, II ,III C) I, IV D) III ,IV

27. A sample containing 0.496 gm of $(NH_4)_2\ C_2O_4\ (MW = 124)$ and inert material was dissolved in water and made strongly alkaline with KOH which converts NH_4^+ into NH_3. The liberated NH_3 was distilled into exactly 50ml of 0.05M H_2SO_4. The excess H_2SO_4 was back titrated with 10ml of 0.1M $NaOH$. The percentage of $(NH_4)_2\ C_2O_4$ in the sample is

A) 40% B) 50% C) 60% D) 75%

28. An electron, practically at rest, is initially accelerated through a potential difference of 100 volts. It then has a de Broglie wavelength $= \lambda_1^0 A$. It then gets retarded through 19 volts and then has a wavelength $= \lambda_2^0 A$. A further retardation through 32 volts changes the wavelength to λ_3. What is $\frac{\lambda_3 - \lambda_2}{\lambda_1}$?

A) $\frac{20}{41}$ B) $\frac{10}{63}$ C) $\frac{20}{63}$ D) $\frac{10}{41}$

For More Join: @IITJEE_Advanced

29. Arrange the following compounds in increasin order of their dipole moment :
HBr, H_2S, NF_3 and $CHCl_3$

A) $NF_3 < HBr < H_2S < CHCl_3$ B) $HBr < H_2S < NF_3 < CHCl_3$

C) $H_2S < HBr < NF_3 < CHCl_3$ D) $CHCl_3 < NF_3 < HBr < H_2S$

30. Consider the following statements:

I) Percentage of σ - bonding in $C-O$ bond follows the sequences $CO < CO_2 < CO_3^{2-}$

II) Relative strength of π - bonding in $BX_3\left(X-F,Cl,Br,I\right)$ follows the sequence

$$BF_3 = BCl_3 > BBr_3 > BI_3$$

III) The correct order of bond lengths (S-O) follows the sequence $SO_3^{2-} > SO_4^{2-}$

IV) bond order follows the sequence $ClO_2^- > ClO_3^- > ClO_4^-$

Using 'T' for "True" and 'F' for 'False' statement in the given sequence, pick the correct set of codes.

A) TFTT B) TFTF C) TTFT D) FTTF

31. When $KMnO_4$ acts as an oxidising agent it may change to MnO_4^{-2}, MnO_2, Mn_2O_3 and Mn^{+2}.

The number of electrons transferred in each case respectively are

A) 4, 3, 1, 5 B) 1,5, 3, 7 C) 1, 3, 4, 5 D) 3, 5, 7, 130

32. Calculate $\Delta_f G^0$ for $\left(NH_4Cl,s\right)$ at 310 K.

$\Delta_f H^0\left(NH_4Cl,s\right) = -314.5kJ/mol, \Delta_r C_p = 0$

$S^0 N_{2(g)} = 192JK^{-1}mol^{-1}$

$S^0{}_{H_{2(g)}} = 130.5JK^{-1}mol^{-1}$

$S^0 Cl_{2(g)} = 233JK^{-1}mol^{-1}$

$S^0{}_{NH_4Cl(s)} = 99.5JK^{-1}mol^{-1}$

All given data at 300K.

A) $-198.56\,kJ/mol$ B) $-426.7\,kJ/mol$ C) $-202.3\,kJ/mol$ D) $-84.5\,kJ/mol$

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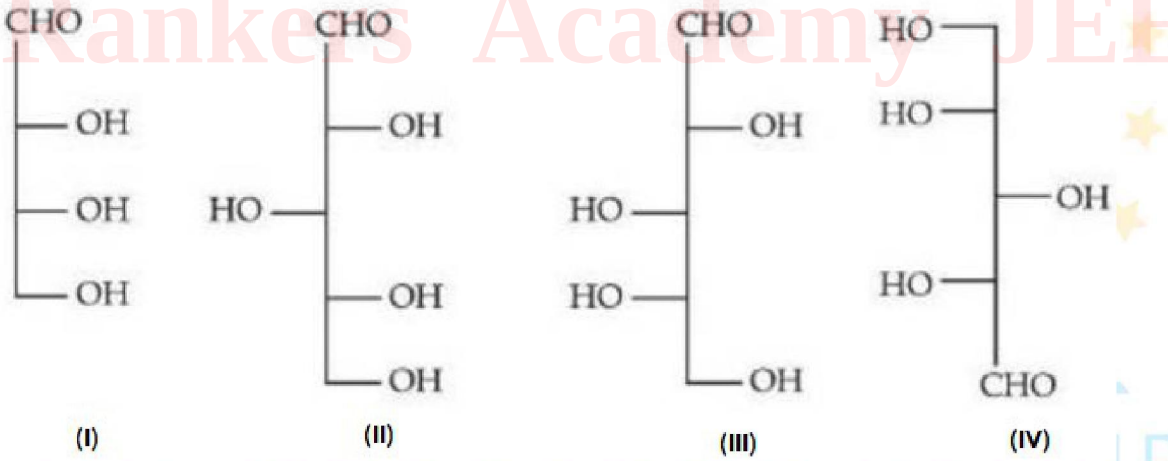
33. When N_2O_5 is heated at certain temperature, it dissociates as $N_2O_5(g) \rightleftharpoons N_2O_3(g)+O_2(g)$ $K_c=2.5$. At the same time N_2O_3 also decomposes as $N_2O_3(g) \rightleftharpoons N_2O(g)+O_2(g)$. If initially 4.0moles of N_2O_5 are taken in 1.0 liter flask and allowed to dissociation, concentration of O_2 at equilibrium is 2.5M. Equilibrium concentration of N_2O_5 is:

A) 10.M B) 1.5M C) 2.166M D) 1.83

34. The enthalpies of neutralization AOH and a strong base BOH by HCl are -12250 cal/mol and $-13000 cal/mol$ respectively. When one mole of HCl is added to a solution containing 1 mole of AOH and 1 mole of BOH, the enthalpy change was -12500 cal/mol. In what ratio is the acid distributed between AOH and BOH respectively.

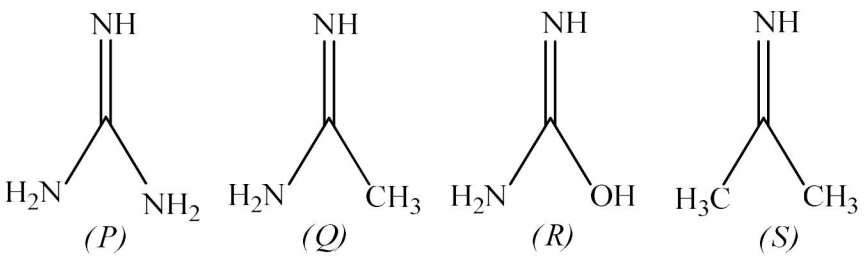
A) 2:1 B) 2:3 C) 1:2 D) 3:2

35. Identify the number of structure/s from the following which can be correlated to D-glyceraldehyde.



A) three B) two C) four D) one

36. Arrange the following in increasing order of basicity:



A) $S<Q<R<P$ B) $S<Q<P<R$ C) $S<R<P<Q$ D) $P<Q<R<S$

For More Join: @IITJEE_Advanced

37. Given below are two statements :

Statement I: One mole of propyne reacts with excess of sodium to liberate half mole of H_2 gas.

Statement II: 4 gr of propyne reacts with $NaNH_2$ to liberate NH_3 gas which occupies 224 mL at STP.

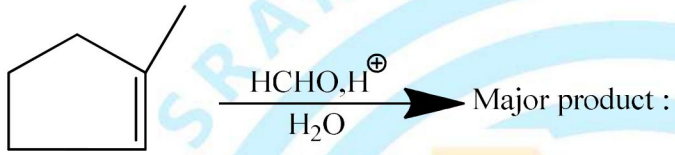
In the light of the above statements, choose the most appropriate answer from the options given below:

A) Statement I is correct but Statement II is incorrect.

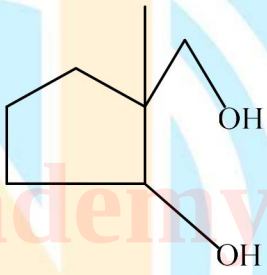
B) Both Statement I and Statement II are incorrect

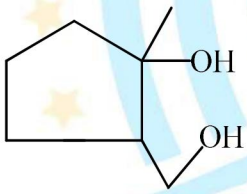
C) Statement I is incorrect but Statement II is correct

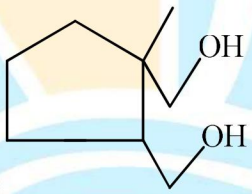
D) Both Statement I and Statement II are correct.

38. 

A) 

B) 

C) 

D) 

39. Match List-I with List-II

List-I		List-II	
A)	$Al^{3+} < Mg^{2+} < Na^{+} < F$	I)	Ionisation Enthalpy
B)	$B < C < O < N$	II)	Metallic character
C)	$B < Al < Mg < K$	III)	Electronegativity
D)	$Si < P < S < Cl$	IV)	Ionic radii

Choose the correct answer from the options given below :

A) A – IV, B – I, C – III, D – II

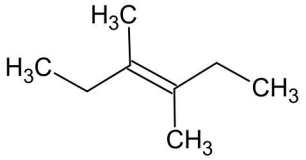
B) A – II, B – III, C – IV, D – I

C) A – IV, B – I, C – II, D – III

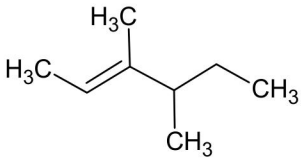
D) A – III, B – IV, C – II, D – I

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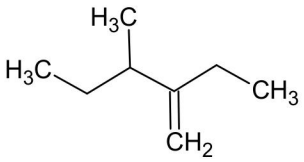
40. Arrange the following alkenes in the correct order of stability.



[X]



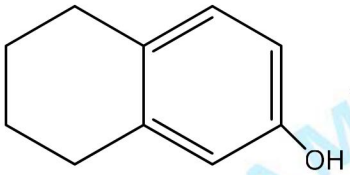
[Y]



[Z]

A) $X > Z > Y$ B) $X > Y > Z$ C) $Z > X > Y$ D) $Y > X > Z$

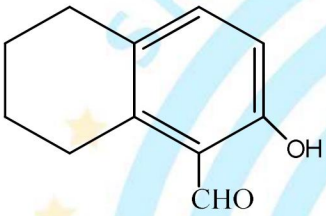
41.



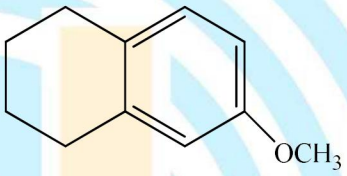
$CHCl_3$+NaOH → A (Formylation) + B (Other Product)

The other product B is:

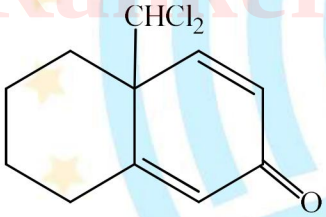
A)



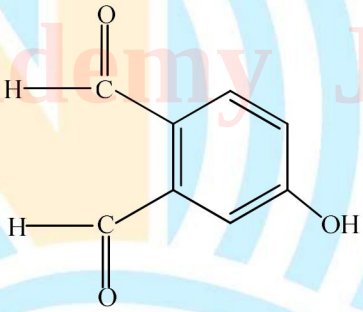
B)



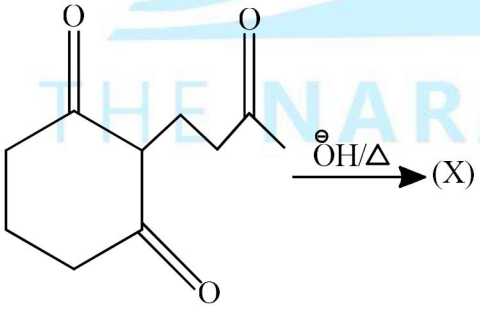
C)



D)

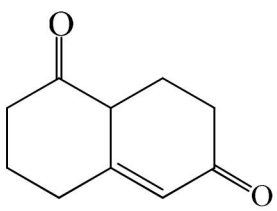


42. What is the product (X) of the following reaction?

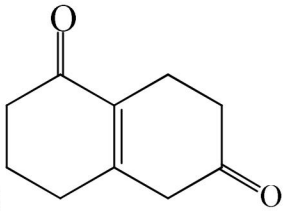


$\overset{\ominus}{O}H/\Delta$ → (X)

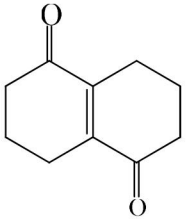
A)



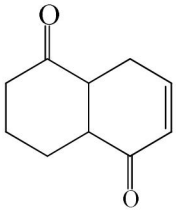
B)



C)

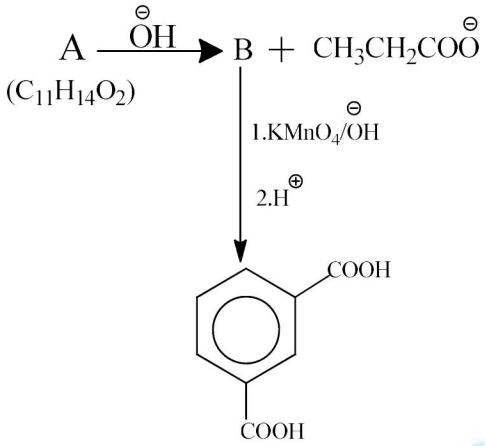


D)

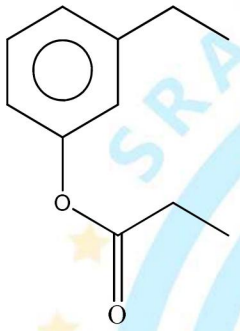


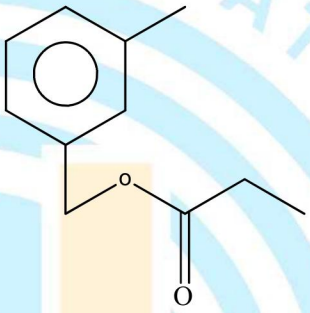
For More Join: @IITJEE_Advanced

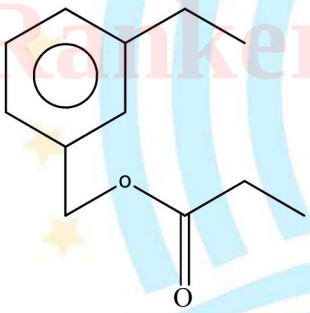
43. Consider the following sequence of reaction

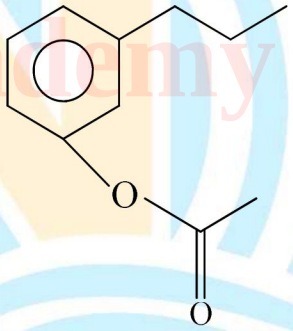


The compound A is:

A) 

B) 

C) 

D) 

44. Given below are two statement :

Statement (I): Nitrogen, sulphur, halogen and phosphorus present in an organic compound are detected by Lassaigne's Test.

Statement (II): The elements present in the compound are converted from covalent form into ionic form by fusing the compound with Magnesium in Lassaigne's test.

In the light of the above statements, choose the correct answer from the options given below :

A) Both Statement I and Statement II are true

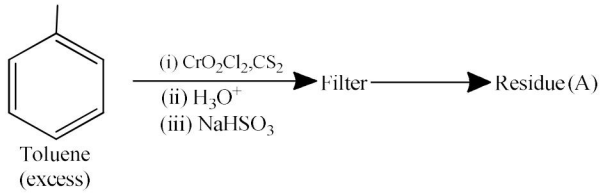
B) Both Statement I and Statement II are false

C) Statement I is true but Statement II is false

D) Statement I is false but Statement II is true

For More Join: @IITJEE_Advanced

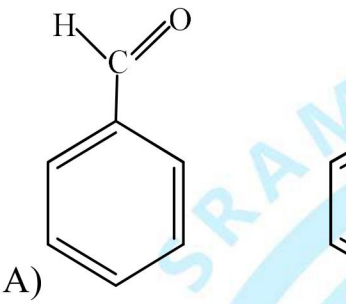
45.

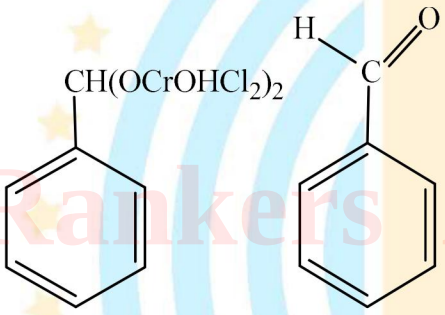


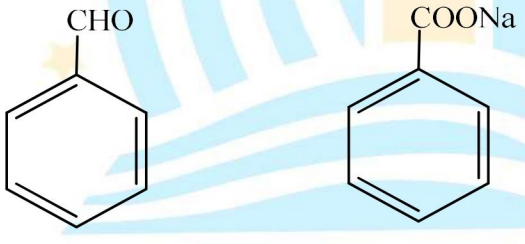
$Residue\ (A) + HCl(dil.) \rightarrow Compound\ (B)$

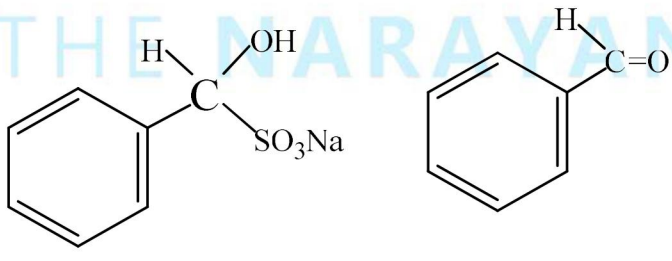
Structure of residue (1) and compound (2) formed respectively is:

(1) (2)

A) 

B) 

C) 

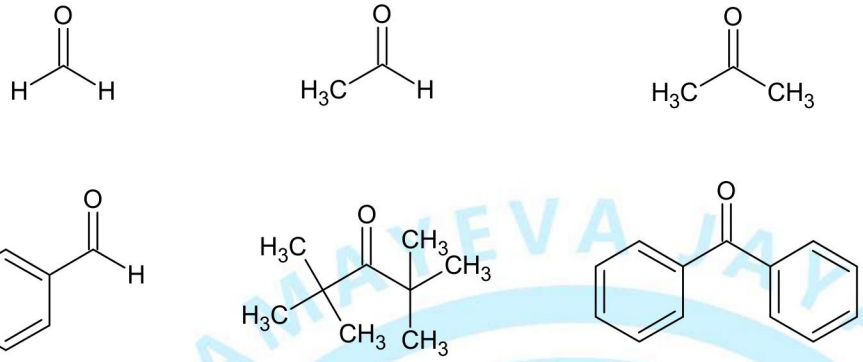
D) 

For More Join: @IITJEE_Advanced

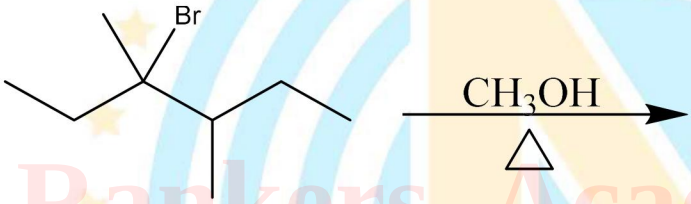
SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

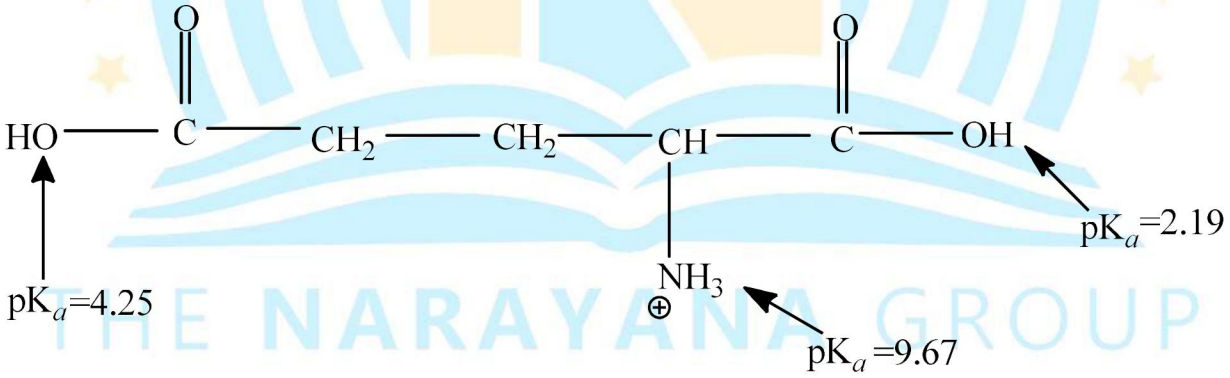
46. Among the given how many of the following can show Cannizzaro reaction?



47. The total number of (including stereo isomers) E_1 products (alkenes) formed in the following reaction is



48. The P^I (Isoelectric point) of the following amino acid is



49. The number of molecules/ions that show linear geometry among the following is ____ .
$SO_2, BeCl_2, CO_2, N_3^-, NO_2, F_2O, XeF_2, NO_2^+, I_3^-, O_3$

50. If the freezing point of a 0.01 molal aqueous solution of a cobalt (III) chloride – ammonia complex (which behaves as a strong electrolyte) is $-0.0558^0 C$, the number of chloride(s) in the coordination sphere of the complex is : $\left[K_f \text{ of water} = 1.86 K \text{ kg mol}^{-1}\right]$

For More Join: @IITJEE_Advanced

MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Let $g(x)=\int\frac{4x^7+6x^5}{\left(1+x^2+3x^6\right)^2}dx$. For $x>0, \lim_{x\to 0} g(x)=0$. If $A=\begin{bmatrix}0 & 0 & 1\\ 5 & 1 & 1\\ \alpha^2 & g(1) & 1\end{bmatrix}$ and $B=adj\left(adj\ A\right)$ is such that $|B|=16$ and $\alpha\in R$, then the value of α^2 is equal to

A) 1 B) 2 C) 3 D) 4

52. Let the length of the latus rectum of an ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1(a>b>0)$ be 15. If its eccentricity is the maximum value of the function $f(t)=\frac{1}{4}-t^2+2t-1$, then the value of $\left(a^2+b^2\right)$ is equal to:

A) 124 B) 144 C) 156 D) 164

53. Let the angles made with the positive x-axis by two straight lines drawn from the point $P(1,2)$ and meeting the line $x+y=5$ at a distance of $\sqrt{\frac{8}{3}}$ from the point P be α_1 and α_2. Then the value of $(\alpha_1+\alpha_2)$ is:

A) $\frac{\pi}{4}$ B) $\frac{\pi}{2}$ C) $\frac{\pi}{3}$ D) $\frac{2\pi}{3}$

54. Let $\vec{a}=\hat{i}+\hat{j}$ and $\vec{b}=\hat{j}+\hat{k}$. Let $\vec{\upsilon}$ be a vector in the plane of the vectors $\vec{a}$ and $\vec{b}$ such that $\vec{\upsilon}.\vec{a}=0$. If the length of the projection of $\vec{\upsilon}$ on the vector $\vec{c}=\hat{i}+2\hat{j}-\hat{k}$ is $\sqrt{6}$, then $|\vec{\upsilon}|^2$ is equal to

A) 108 B) 72 C) 216 D) 36

55. Let a_1,a_2,a_3,a_4 be an A.P . of four terms such that each term of A.P. and its common difference d are integers. If $a_1+a_2+a_3+a_4=24$ and $a_1.a_2.a_3.a_4+d^4=961$, then the largest term of the A.P. is equal to:

A) 7 B) 9 C) 11 D) 13

For More Join: @IITJEE_Advanced

56. Let the image of the parabola $y^2 = 4x$ in the line $x + y = 2$ be $(x-a)^2 = b(y-c)$, where $b < 0$

Then $a - b + c$ is equal to:

A) 4 B) 6 C) 8 D) 10

57. $\left(\frac{1}{3}+\frac{4}{7}\right)+\left(\frac{1}{3^2}+\frac{1}{3}\times\frac{4}{7}+\frac{4^2}{7^2}\right)+\left(\frac{1}{3^3}+\frac{1}{3^2}\times\frac{4}{7}+\frac{1}{3}\times\frac{4^2}{7^2}+\frac{4^3}{7^3}\right)+.....$ upto infinite terms equal to –

A) $\frac{5}{2}$ B) $\frac{7}{4}$ C) $\frac{4}{3}$ D) $\frac{6}{5}$

58. Let $A = [a_{ij}]$ and $B = [b_{ij}]$ be two square matrices of order 3 such that $b_{ij} = 2^{i+j-2}a_{ij}$ and $\det(B) = 32$. Then the value of $64.\det(adj(A))$ is:

A) 4 B) 8 C) 12 D) 16

59. The letters of the word "BHARAT" are written in all possible ways with or without meaning, and these words are arranged as in a dictionary. The rank of the word "BHARAT" is

A) 146 B) 147 C) 148 D) 149

60. The largest value of n, for which 45^n divides 80!, is:

A) 17 B) 18 C) 19 D) 20

61. The sum of all values of α, for which the shortest distance between the lines

$\frac{x-1}{\alpha} = \frac{y-2}{-2} = \frac{z}{-\alpha}$ and $\frac{x-1}{\alpha} = \frac{y-1}{1} = \frac{z-1}{\alpha}$ is $\frac{1}{\sqrt{2}}$ is:

A) 6 B) –6 C) 4 D) –4

62. Let $g(\lambda)$ denote the area of the region in the first quadrant bounded by

$x = 0, x = 1$ $y = \sqrt{x}$ and

$y = |\lambda x - 4| - |2 - \lambda x| + \lambda x^2$. Then $(g(0) + g(2))$ is equal to:

A) $\frac{8}{3}$ B) $\frac{10}{3}$ C) 4 D) $\frac{14}{3}$

For More Join: @IITJEE_Advanced

63. If the domain of the function $f(x)=\cos^{-1}\left(\frac{2}{x^2-x-1}\right)$ is $(-\infty,a]\cup[d,\infty)$, then $a+d$ is equal to:

A) 0 B) 1 C) 2 D) 3

64. Let $X=\{x\in\mathbb{N}:1\le x\le 13\}$ and for some $a,b\in\mathbb{R}, Y=\{ax+b:x\in X\}$. If the mean and variance of the elements of Y are 20 and 126 respectivley, then the sum of all possible values of b is:

A) 20 B) 30 C) 40 D) 50

65. Consider the following three statements for the function $f:\mathbb{R}\to\mathbb{R}$ defined by $f(x)=|e^x-1|-|x|$:

(I) f is differentiable at $x=0$ (II) f is increasing in $(0,\infty)$

(III) f is decreasing in $(-\infty,0)$ Then

A) All (I), (II) and (III) are TRUE B) Only (I) and (II) are TRUE

C) Only (II) and (III) are TRUE D) Only (I) and (III) are TRUE

66. Let $y=y(x)$ be differentiable function in the interval $(0,\infty)$ such that $y(1)=1$ and $\lim\limits_{t\to x}\left(\frac{t^2y(x)-x^2y(t)}{x-t}\right)=2$ for each $x>0$. Then $3y(2)$ is equal to:

A) 15 B) 17 C) 19 D) 21

67. Let f be a function such that $2f(x)+3f\left(\frac{10}{x}\right)=4x$ for all $x>0$. Then $5\left(f(2)-f(5)\right)$ is equal to:

A) 40 B) 50 C) 60 D) 70

68. The smallest positive integral value of k, for which all the roots of $x^4-kx^2+16=0$ are real and distinct, is equal to:

A) 7 B) 8 C) 9 D) 10

For More Join: @IITJEE_Advanced

69. Let $\vec{a}=\hat{i}+\hat{j}-\hat{k}$ and $\vec{b}=2\hat{i}-\hat{j}+2\hat{k}$. If $\vec{c}$ is a vector such that $\left(\vec{a}+2\vec{b}\right)\times\vec{c}=\vec{0}$ and $\left(\vec{a}+\vec{b}\right).\vec{c}=18$, then $|\vec{c}|^2$ is equal to :

A) 25 B) 35 C) 45 D) 50

70. If the function $f(x)=\begin{cases}\dfrac{1-\cos(bx)}{x^2}, & x<0\\ a, & x=0\\ \dfrac{\tan x-\sin x}{x^3}, & x>0\end{cases}$ is continuous at $x=0$ then a^2+b^2 is equal to

A) $\frac{5}{4}$ B) $\frac{3}{2}$ C) 2 D) $\frac{1}{4}$

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

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71. If $f(x)$ satisfies the relation $f(x)=1+\int_0^1\left(x+y^2\right)f(y)\,dy$, then the value of $f(1)$ is equal to

72. Let (h,k) lie on the circle $C:x^2+y^2=9$ and the point $(3h-1,2k+1)$ lie on a ellipse with eccentricity e for all such (h,k). Then the value of $9e^2$ is equal to _____.

73. Let $z=(1+i)(1+2i)(1+3i).....(1+ni)$, where $i=\sqrt{-1}$. If $|z|^2=1700$, then n is equal to _____.

74. Let S be a set of 4 elements and $P(S)$ denote the power set of S. Let E be an event of choosing an ordered pair (A,B) from the set $P(S)\times P(S)$ such that $A\cup B=S$. If the probability of the event E is $\frac{3^p}{2^q}$ where $p,q\in\mathbb{N}$, then $p+q$ is equal to _____.

75. The number of elements in the set $\left\{x\in[0,\pi]:\tan\left(x+\frac{2\pi}{3}\right)=\tan\left(x+\frac{\pi}{3}\right)\tan x\tan\left(x-\frac{\pi}{3}\right)\right\}$ is __

For More Join: @IITJEE_Advanced